

Testing for **Cations** Using Sodium Hydroxide

Definitions:

Ion: charged particle.

Cations: positively charged particles.

Anions: negatively charged particles.



CATS ARE GOOD
(POSITIVE)



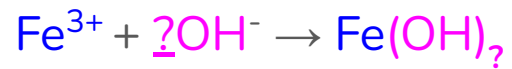
ONIONS
ARE
BAD
(NEGATIVE)

What Did You See?

Substance	Positive ion solution	Sodium hydroxide
	Iron(II), Fe ²⁺	
	Iron(III), Fe ³⁺	
	Copper(II), Cu ²⁺	
	Aluminium, Al ³⁺	
	Calcium, Ca ²⁺	
	Magnesium, Mg ²⁺	
	Ammonium, NH ₄ ⁺	

Table 1 Sample observations

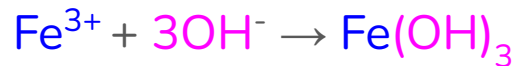
Ionic Equations



$$+3 - (?) = 0$$

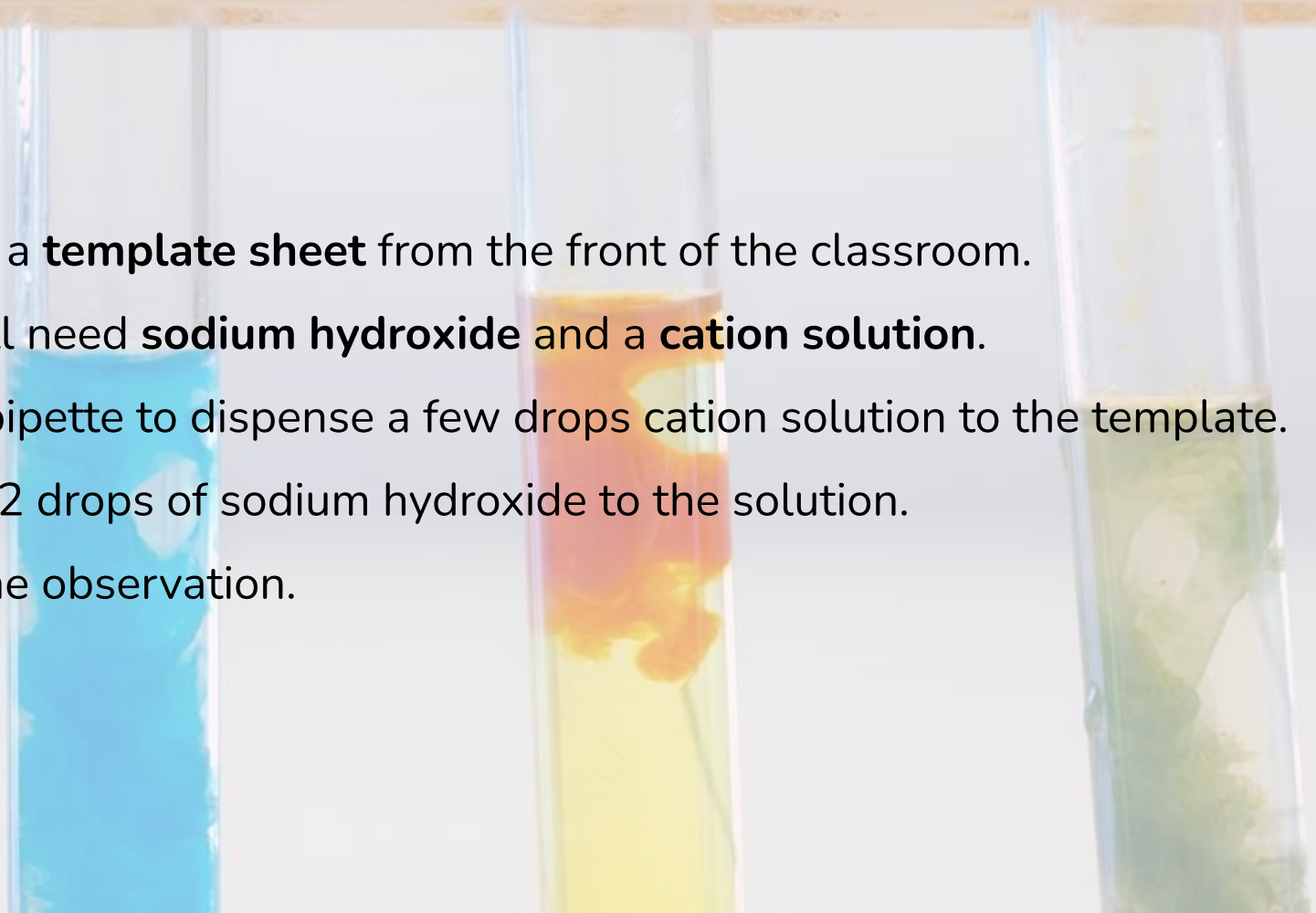
$$+3 - (3) = 0$$

So we need 3 hydroxide ions!!



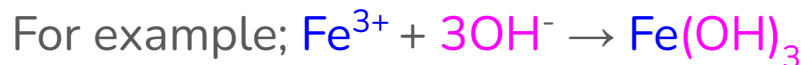
Method:

1. Collect a **template sheet** from the front of the classroom.
2. You will need **sodium hydroxide** and a **cation solution**.
3. Use a pipette to dispense a few drops cation solution to the template.
4. Add 1-2 drops of sodium hydroxide to the solution.
5. Note the observation.



What is happening?

- Hydroxide ions in NaOH are reacting with the cations to form new hydroxides.



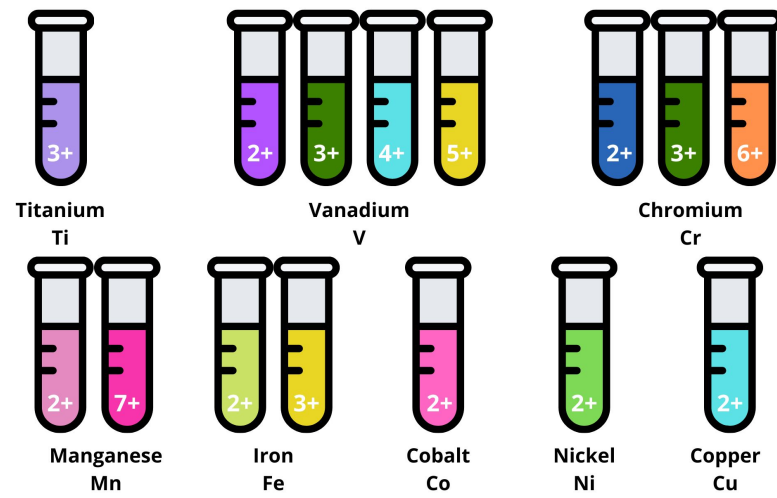
*This is called an **ionic equation** it shows all the **relevant stuff!!**

- In these reactions a solid forms that does not dissolve.
- That's why we can see it!

What is happening?

- For this experiment we used **transition metals** → form *pretty colour ions!!*
- This is because when we remove electrons they have extra special properties.

Transition Metal Ion Colors in Aqueous Solution



What Did You See?

Substance	Positive ion solution	Sodium hydroxide
Iron Sulfate	Iron(II), Fe ²⁺	
Iron Nitrate	Iron(III), Fe ³⁺	
Copper Sulfate	Copper(II), Cu ²⁺	
Aluminium Nitrate	Aluminium, Al ³⁺	
Calcium Chloride	Calcium, Ca ²⁺	
Magnesium Chloride	Magnesium, Mg ²⁺	
Ammonia Chloride	Ammonium, NH ₄ ⁺	

Table 1 Sample observations